

Hartshorne Chapter 1

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Section 1.3

Question 3.1

a)

By the result of Question 1.1, if $Y \subseteq \mathbf{A}^2$ is a conic then $I(Y)$ is isomorphic to either $k[x]$ or $k[x, y]/\langle xy - 1 \rangle$. It is clear that $k[x]$ is the coordinate ring of $\mathbf{A}^1$, and we get isomorphism in the first case. On the other hand, note that $Z(xy - 1) \subseteq \mathbf{A}^2$ is given by $\{(x, \frac{1}{x}) \mid x \in k \setminus \{0\}\}$ such that it is isomorphic to $\mathbf{A}^1 \setminus \{0\}$ via the map $(x, \frac{1}{x}) \mapsto x$. $\square$

b)

Suppose $U \subset \mathbf{A}^1$ is any open set. Let $z \in \mathbf{A}^1 \setminus U$ be any point not contained in U , such that the rational function $\frac{1}{x-z}$ is regular on U . But then $\mathcal{O}(U)$ has a nonconstant invertible function, such that $\mathcal{O}(U)$ cannot be isomorphic to $\mathcal{O}(\mathbf{A}^1) \simeq k$. Thus $\mathbf{A}^1$ is never isomorphic to any proper open subset of itself. $\square$

c)

How do you do this

d)

By the result of Question 3.7, any two curves in $\mathbf{P}^2$ has nonempty intersection. But then that is not the case in $\mathbf{A}^2$ (choose any two parallel lines). Hence they cannot be isomorphic, as any homeomorphism preserves the dimension of sets. $\square$

e)

If a projective variety X is isomorphic to an affine variety Y , the definition of isomorphism necessarily gives a bijection between globally regular functions such that $A(Y)$ must be isomorphic to $\mathcal{O}(Y) \simeq k$. But then by Nullstellensatz, the ideal of Y is maximal and hence Y is a singular point. $\square$

Question 3.2

a)

That φ defines a homomorphism is clear, both inherit the cofinite topology. Moreover by Lemma 3.6 it is also immediate that it defines a morphism.

Consider the regular function on $\mathbf{A}^1 \setminus \{0\}$ given by $t \mapsto \frac{1}{t}$. Then if φ were to be an isomorphism, its pullback $(t^2, t^3) \mapsto \frac{1}{t}$ on $\varphi(\mathbf{A}^1) \setminus \{(0, 0)\}$ would have to be regular as well. But then there exists polynomials f, g such that $\frac{f(t^2, t^3)}{g(t^2, t^3)} = \frac{1}{t}$, where g is a polynomial that vanishes nowhere on $\varphi(\mathbf{A}^1)$ except potentially at $(0, 0)$. But then it can only be equal to ct or c ; and the first is impossible as it is a polynomial in t^2, t^3 and the second is impossible as that would imply the polynomial f is the rational function $\frac{1}{t}$. $\square$

b)

Suppose that $t^p = t'^p$. Then $t^p - t'^p = (t - t')^p = 0$ such that φ is injective. As the base field is algebraically complete, that φ is surjective is also clear. By the same reasoning as in (a), φ defines a homeomorphism as both inherit the cofinite topology. Moreover by Lemma 3.6 it is also immediate that it defines a morphism.

The same proof as done in (a) can be repeated to show that the regular function $t \mapsto \frac{1}{t}$ cannot pull back to a regular function on $\varphi(\mathbf{A}^1)$. Hence it is not an isomorphism. $\square$

Question 3.3

a)

In Y , let (U, f) be an element of $\mathcal{O}_{\varphi(P), Y}$. Then by definition it pulls back to a regular function $(\varphi^{-1}(U), f \circ \varphi)$. Hence we get a well-defined map $\varphi_P^* : \mathcal{O}_{\varphi(P), Y} \rightarrow \mathcal{O}_{P, X}$.

That it defines a ring homomorphism is easy to check. $\square$

b)

If φ is an isomorphism, that φ_P^* defines an isomorphism of rings for all $P \in X$ is clear.

Conversely suppose that φ_P^* defines an isomorphism of rings for all $P \in X$. Then φ preserves regular maps in both directions. But then it must also preserve the domain of every regular map, i.e. the open sets. Hence φ is an isomorphism. $\square$

c)

Suppose that $(U, f) \in \mathcal{O}_{\varphi(P), Y}$ such that $\varphi_P^*(f) = 0$. Then we get $(\varphi^{-1}(U), f \circ \varphi)$ in X . But then $\varphi(X)$ is dense in Y such that $\varphi^{-1}(U)$ is a nonempty open set in X . Hence $(\varphi^{-1}(U), f \circ \varphi)$ is equivalent to $(X, 0)$, such that φ_P^* is injective for all $P \in X$. $\square$

Question 3.4

The result of Question 2.12 gives us that the d -uple embedding defines a homeomorphism. Moreover by considering the corresponding ring homomorphism $\theta : k[y_0, \dots, y_N] \rightarrow k[x_0, \dots, x_n]$ and the induced map between $K(\mathbf{P}^N)$ and $K(\mathbf{P}^n)$ it is clear to see that the d -uple embedding is a morphism.

Now we give an explicit form for the inverse of ρ_d piecewise. Consider the open set of $\mathbf{P}^N$ given by $x_i \neq 0$ (WLOG set $i = 0$). Then notice that in $\mathbf{P}^n$, we have that $(x_0 : \dots : x_n) = (x_0^d : \dots : x_0^{d-1}x_n)$, such that each coordinate corresponds to a d -monomial and the inverse map can be explicitly defined. But then ρ_d^{-1} is a morphism when restricted to each U_i , such that it patches to a morphism on $\mathbf{P}^N$ as regular functions also patch. $\square$

Question 3.5

Suppose that $Z(f) = H \subseteq \mathbf{P}^n$ is a hyperplane. Then let $\deg f = d$, and consider the d -uple embedding $\mathbf{P}^n \hookrightarrow \mathbf{P}^N$ as given in Question 3.4 above. Then f must necessarily be mapped to a linear polynomial in $\mathbf{P}^N$, such that it defines a hyperplane. But then the complement of a hyperplane in projective space is isomorphic to affine space, such that the image of $\mathbf{P}^n \setminus H$ in $\mathbf{P}^N$ is isomorphic to a subset of an affine variety, i.e. an affine variety. $\square$

Question 3.6

Suppose that $\frac{f}{g} \in \mathcal{O}(X)$. Then g can vanish nowhere on $\mathbf{A}^2$ except potentially at the origin. But then so must $g(c, y)$ for any constant c , such that the algebraic closure requires that $g(c, y)$ be a constant. By symmetry that implies that $g(x, c)$ so must be one as well such that g is a constant function. Hence we get an surjection $k[x, y] \twoheadrightarrow \mathcal{O}(X)$, and it is easy to see that it is injective.

Now, notice that the affine coordinate ring of $\mathbf{A}^2$ is also $k[x, y]$; such that by Prop 3.5 X would have to be isomorphic to $\mathbf{A}^2$ were it an affine variety. But then by considering the explicit construction of the bijection given in Prop 3.5, the identity map (which is an isomorphism) on $k[x, y]$ must correspond to the inclusion map $X \hookrightarrow \mathbf{A}^2$, which is not an isomorphism. Hence we get a contradiction. $\square$

Question 3.7

a)

Follows as a corollary of (b).

b)

Suppose that $Y \subseteq \mathbf{P}^n$ is a projective variety of dimension at least 1, and $H \subseteq \mathbf{P}^n$ is a hypersurface. Then suppose that $Y \cap H = \emptyset$. Then $Y = Y \setminus H$ is isomorphic to an affine variety by the result of Q3.5, such that Y is a singular point by the result of Q3.1e. But then the dimension of Y is necessarily 0, which is a contradiction. $\square$

Question 3.8

How do you do this #2

Question 3.9

Let the 2-uple embedding be given explicitly by $(x : y) \mapsto (x^2 : y^2 : xy)$. Then the irreducible polynomial $xy - z^2$ is contained inside $I(Y)$, such that $I(Y) = \langle xy - z^2 \rangle$ as $\dim(Y) = \dim(\mathbf{P}^1) = 1$.

Hence $S(Y) \simeq k[x, y, \sqrt{xy}]$. Now if it were to be isomorphic to $k[x, y]$ via some ϕ , notice that $\phi(x)\phi(y) - \phi(\sqrt{xy})^2 = 0$ necessarily. But then $\phi(x), \phi(y)$ are both still irreducible elements such that $\phi(x) = \phi(y) = \phi(\sqrt{xy})$, which is a contradiction. Hence they are not isomorphic. $\square$

Question 3.10

That $\varphi_{X'}$ is a continuous map is clear.

Now suppose that $(U, \frac{f}{g})$ is some locally regular map in Y' . Then $U = C \cap \overline{U}$ for some closed $C \subset Y$ and open $\overline{U} \subseteq Y$. Then $\hat{U} := Y \setminus Z(g)$ is open in Y , such that $\frac{f}{g}$ extends to a well-defined regular function on it. But then $\frac{f}{g}$ pulls back to a regular function on $\varphi^{-1}(\hat{U})$, such that it must on $\varphi_{X'}^{-1}(U)$ as well by the commutativity of the diagram. $\square$

Question 3.11

Question 3.12

Question 3.13

Question 3.14

Question 3.15

Question 3.16

Question 3.17

Question 3.18

Question 3.19

Question 3.20

Question 3.21